

Aim:

To write a Java program to convert a Roman numeral into its equivalent integer value.

Algorithm:

Start the program.

Read a Roman numeral from the user.

Convert the input to uppercase.

Initialize total = 0 and previous = 0.

Traverse the Roman numeral from right to left.

Assign the corresponding integer value to each Roman symbol.

If the current value is less than the previous value, subtract it from total; otherwise, add it.

Update previous with the current value.

Display the integer value.

Stop the program.

Steps:

Import the Scanner class to read input.

Create a class named Roman.

Read a Roman numeral from the user.

Use a switch statement to convert each symbol into its integer value.

Use a loop to calculate the result.

Display the equivalent integer value.

Code:

```
package basics;
import java.util.Scanner;

public class Roman {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Roman numeral: ");
        String s = sc.nextLine().toUpperCase();

        int total = 0;
        int previous = 0;

        for (int i = s.length() - 1; i >= 0; i--) {
            int value = 0;

            switch (s.charAt(i)) {
                case 'I': value = 1; break;
                case 'V': value = 5; break;
                case 'X': value = 10; break;
```

```

■          case 'L': value = 50; break;
■          case 'C': value = 100; break;
■          case 'D': value = 500; break;
■          case 'M': value = 1000; break;
■          default:
■              System.out.println("Invalid Roman numeral");
■              sc.close();
■              return;
■      }

■      if (value < previous) {
■          total -= value;
■      } else {
■          total += value;
■      }

■      previous = value;
■  }

■      System.out.println("Integer value: " + total);
■      sc.close();
■  }
■}

```

UML Diagram:

```

+-----+
| Roman |
+-----+

| - total : int |
| - previous : int |
| - value : int |
| - s : String |
+-----+

| + main(args : String[]) : void |
+-----+

```

Output: